GF/PED Manual

Wilson GF analysis, internal-coordinate force constants and potential-energy distributions in ORACLE

ORACLE Development Notes

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Abstract

`GF/PED` is the `ORACLE` workflow that transforms a Cartesian harmonic Hessian into an internal-coordinate Wilson GF problem, solves the harmonic vibrational eigenproblem and reports frequencies, internal force constants, normal modes and potential-energy distributions (PEDs). The production path uses a frozen non-redundant `GICForge` definition, so vibrational analysis, Gaussian input generation and structural refinement can share the same coordinate model. This manual describes the theory, input contracts, command line and GUI workflows, Pulay-style scaling, output files and validation rules.

Contents

1 Purpose

The GF/PED module answers a specific question: given a Cartesian Hessian and a non-redundant internal-coordinate basis, what are the harmonic frequencies, normal modes, internal force constants and coordinate contributions in that basis? It is deliberately separate from VPT2/VCI. VPT2/VCI works in normal-coordinate quartic force fields; GF/PED works with a Cartesian Hessian, a Wilson B matrix and a non-redundant coordinate model.

The preferred production workflow is `gic-gf`. It reads a frozen `GICForge` schema and a Gaussian FCHK Hessian adapter, evaluates the same GICs on the Hessian geometry or on an optional external geometry, transforms the Cartesian Hessian to the GIC basis and solves Wilson GF. A simpler diagnostic workflow, `gf`, reads an FCHK file and generates a ORACLE GIC basis on the fly; it is useful for quick checks but does not provide the same cross-module coordinate identity as a frozen `GICForge` schema.

2 Theoretical Background

Let x be the $3N$-dimensional Cartesian coordinate vector, F_x the Cartesian harmonic Hessian in E_h/bohr^2, and m_i the atomic masses in amu. For a non-redundant internal coordinate vector q, the linearized relation between Cartesian and internal displacements is

$$dq = B\,dx,$$

where $B = \partial q/\partial x$ is the Wilson B matrix. The kinetic-energy matrix in internal coordinates is

$$G = B\,M^{-1}\,B^T,$$

where M^{-1} is the diagonal Cartesian inverse-mass matrix with each atomic mass repeated for x, y, z.

The internal-coordinate Cartesian backtransform used by the code is

$$A = M^{-1}B^TG^+,$$

where G^+ is the numerical inverse or pseudoinverse of G. The internal force-constant matrix is then

$$F_q = A^TF_xA.$$

The matrix is explicitly symmetrized before the GF solution. If Pulay-style diagonal scaling factors s_i are supplied, the scaled internal Hessian is

$$(F_q^{\text{scaled}})_{ij} = (F_q)_{ij}\sqrt{s_is_j}.$$

This preserves symmetry and applies off-diagonal scaling as the geometric mean of the two diagonal factors.

The Wilson GF eigenproblem is solved in symmetric form. Since G is positive definite for a valid non-redundant basis, the code diagonalizes

$$G^{1/2}F_qG^{1/2}.$$

The eigenvalues are converted to signed harmonic frequencies in cm^{-1}. Negative eigenvalues are retained as negative frequencies and therefore expose transition-state or instability directions rather than being silently clipped.

The internal normal modes are back-expressed in the GIC basis, and the potential-energy distribution for coordinate i in mode k is evaluated from the mode vector l_k as

$$\mathrm{PED}_{ik} = 100\,\frac{|l_{ik}(F_q l_k)_i|}{\sum_j |l_{jk}(F_q l_k)_j|}.$$

The absolute-value normalization makes the PED table robust for coupled modes with positive and negative cross terms, while retaining the relative coordinate contribution pattern.

2.1 Large-amplitude GIC subspaces

GF/PED does not identify torsions by projecting Cartesian normal modes. The large-amplitude coordinates are already part of the frozen non-redundant GIC basis supplied by NEO. GF/PED therefore treats torsions, ring puckering, butterfly, out-of-plane, linear-bend and special fragment/centre coordinates as ordinary rows and columns of the same internal-coordinate F and G matrices.

For diagnostics and later anharmonic workflows, GF/PED extracts the corresponding GIC subsets and solves mono- or poly-dimensional Wilson GF problems on their direct F_{SS} and G_{SS} subblocks. No new coordinates are generated and no Cartesian projection is performed. The report also gives the largest F and G couplings between each selected subspace and the remaining coordinates, plus the total large-amplitude PED contribution to every GF mode. These diagnostics decide whether a one-dimensional, coupled multidimensional, DVR or perturbative treatment is appropriate downstream. The operational isolation test uses both matrices: a large-amplitude subspace is independent only when the relative off-subspace coupling is small in F and in G. MATRIX therefore reports the two values separately and also exports $\max(F_{\mathrm{rel}}, G_{\mathrm{rel}})$ as the combined `FG` diagnostic used by downstream rotor/DVR planning.

2.2 Normal-mode and local-mode classes

The GF output now records a zero-order anharmonic model for each GIC. The classification is intentionally chemical and conservative. Coordinates whose diagonal anharmonicity is expected to be smaller than the harmonic off-diagonal couplings remain in the coupled harmonic normal-mode space and are the natural input for VPT2/VCI. Coordinates whose diagonal anharmonicity is expected to dominate their harmonic coupling are tagged as local-mode candidates for VSCF or one-/low-dimensional DVR, with the residual coupling to normal coordinates treated perturbatively when the F/G diagnostics support that approximation.

The default local classes are unsymmetrized X–H stretches selected by NEO, out-of-plane coordinates, ring puckering/ring-butterfly coordinates and torsions around single bonds. Torsional assignment uses the frozen topology and bond-order threshold: high bond-order central bonds are kept in the normal-mode/VPT2 class, while unknown bond orders are marked pending rather than silently converted to hindered rotors. X–H stretches are not forced into the local class by GF itself; the choice belongs to NEO through `XH_STRETCH_POLICY`, `LOCAL_XH_BONDS` and `LOCAL_XH_CLASSES`. This lets the user keep all X–H stretches symmetry-adapted, keep all of them local, or keep only selected `XH`, `XH2` or `XH3` geminal groups local.

The reference mass-metric data used to plan the DVR branch are not recomputed from a separate rotor geometry algorithm. Since the GF calculation already owns the Wilson G matrix in the final non-redundant GIC basis, MATRIX inverts that global equilibrium matrix once and keeps G^{-1} as a reference covariant metric. The DVR plan records $(G^{-1})_{ii}$ for each large-amplitude coordinate. For angular coordinates this is the effective reduced moment in the current GIC metric. For coupled large-amplitude treatments MATRIX stores the full selected block G^{-1}_{SS}, not only its diagonal, because the multidimensional DVR coordinate coupling starts from the off-diagonal metric elements. This exported block is not the final kinetic operator for a large-amplitude DVR. Along a scan or grid the B matrix, $G(Q)$, its inverse $g(Q) = G(Q)^{-1}$, and $\det g(Q)$ generally depend on the large-amplitude coordinates Q. Because $G(Q)$ does not commute with the momentum operators, a rigorous quantum DVR must use a Hermitian curvilinear operator, for example the Laplace–Beltrami/Podolsky form, including the metric determinant and the associated pseudopotential or derivative terms. A production DVR must therefore either recompute the selected metric and determinant at each grid geometry or use an explicit derivative expansion of the metric. The GF export is the equilibrium reference used for planning, diagnostics, one-dimensional bootstrap estimates and consistency checks. Every torsional, ring-puckering, butterfly, out-of-plane, linear-bend and special coordinate is retained as a candidate. The local one-coordinate GF frequency marks whether the candidate is active; by default the CLI uses `250 cm-1`, configurable with `-large-amplitude-frequency-cutoff-cm`. Coordinates above the cutoff are written as excluded candidates rather than removed from the diagnostics.

For ordinary torsions, GF/PED also prepares a DVR plan from the frozen topology and synthon information. The central bond is taken from the torsional primitive, high-order central bonds are excluded from hindered-rotor treatment, and the periodicity is inferred from the equivalence classes of the substituents around the central bond. If the coordinate is isolated according to the combined F/G coupling diagnostic, the local curvature gives the one-term Fourier estimate

$$V(\phi) = A\,[1 - \cos(n(\phi - \phi_0))], \qquad A = F_{\phi\phi}/n^2,$$

with n the inferred periodicity and ϕ_0 the current minimum. If the combined coupling is too large, MATRIX writes a coupled DVR recommendation instead of a one-dimensional barrier.

3 Input Contracts

3.1 Cartesian Hessian

The canonical internal object is `oracle_gf.HessianInput`. It stores atomic numbers, Cartesian coordinates in bohr, atomic masses in amu, a full symmetric Cartesian Hessian in E_h/bohr^2, optional harmonic frequencies from the source adapter and provenance metadata. The preferred full file adapter is Gaussian FCHK/FCH. Gaussian log files can also be promoted when the Cartesian force-constant matrix is printed in the output.

The FCHK file must contain:

- atomic numbers;
- current Cartesian coordinates;
- atomic masses;
- Cartesian force constants;
- harmonic frequencies when available.

The Gaussian log adapter reads only normalized QM output data: the archive geometry, printed Cartesian Hessian, harmonic frequencies and printed normal coordinates when available. It writes the shared `#CARTESIAN_HESSIAN` and `#NORMAL_MODES` sections. GF/PED itself does not parse Gaussian text, does not read a Z-matrix and does not import a ReadAllGIC block from a `.gjf` file.

3.2 Frozen GIC Definition

The production branch reads a `oracle.gic.definition.v1` JSON schema created by `gic-define`. The schema contains primitive coordinates, the GIC coefficient matrix, labels, names, irreducible representations, point group, symmetrization flag and the Gaussian-readable GIC block. GF/PED evaluates that frozen definition; it does not rerun topology perception, pruning or symmetry assignment.

The atom count in the schema and in the FCHK input must agree. If an optional geometry is supplied for the B matrix, it must describe the same atom ordering and the same topology. Coordinates supplied through `-geometry` are in angstrom and are converted internally to bohr before B-matrix evaluation.

4 Command-Line Workflows

4.1 Quick FCHK workflow

The quick workflow builds a ORACLE GIC basis directly from the FCHK geometry:

```
python -m matrix gf \
  --fchk working/gauin.fchk \
  --out working/gf_ped_report.txt \
  --csv-dir working/gf_csv
```

This branch is useful for diagnostics. Because the GICs are generated inside the GF run, it should not be used when the analysis must be identical to a previous Gaussian input, SEfit/MORPHEUS refinement or GICForge manifest.

4.2 Frozen-GIC production workflow

The production workflow first freezes the coordinate model and then runs GF/PED:

```
python -m matrix gic-define \
  --geometry parent.xyz \
  --out working/gic_definition.json \
  --gaussian-out working/gauin

python -m matrix gic-gf \
  --schema working/gic_definition.json \
  --fchk working/gauin.fchk \
  --out working/gic_gf_ped_report.txt \
  --csv-dir working/gic_gf_csv
```

If the B matrix must be evaluated on a geometry different from the FCHK geometry, pass:

```
python -m matrix gic-gf \
  --schema working/gic_definition.json \
  --fchk working/gauin.fchk \
  --geometry working/current.xyz \
  --out working/gic_gf_ped_report.txt
```

This is useful for controlled diagnostics, but production vibrational analysis should normally evaluate the B matrix on the same equilibrium geometry used for the Hessian.

4.3 Gaussian ReadAllGIC log workflow

For a Gaussian job run from MATRIX-generated symmetrized GICs, keep the coordinate definition and the QM result separate, but start the post-run analysis from the optimized Gaussian log. This gives MATRIX the final geometry, not the initial geometry from the `.gjf` input. The GICs are rebuilt and stored by NEO/MATRIX from that final log geometry; the same Gaussian log then contributes Hessian and normal-mode data:

```
matrix link preprocess pyrrole.log pyrrole.xyzin --source-kind gaussian
matrix validate pyrrole.xyzin
matrix neo build pyrrole.xyzin --symmetrize
matrix gaussian promote-log-hessian pyrrole.log pyrrole.xyzin
matrix gf --xyzin pyrrole.xyzin --symmetry-blocks
```

The `pyrrole.gjf` ReadAllGIC block is therefore not treated as an input GIC definition, and its initial Cartesian/Z-matrix geometry is not used for the post-optimization GF/PED analysis. This prevents GF/PED from depending on coordinates that might have been edited, reordered or generated by a different Gaussian-side tool. The only accepted GIC source is the MATRIX `#GIC` section produced by NEO from the final log geometry.
Gaussian log Hessians are stored after Gaussian restores the axes to the archive/original frame. The MATRIX log adapter uses that archive geometry for `#CARTESIAN_HESSIAN`, not the last `Standard orientation` table. When Gaussian prints normal coordinates, `promote-log-hessian` also writes `#NORMAL_MODES` and rotates those printed mode vectors to the archive/original frame when the geometry fit is unambiguous. Add `-no-normal-modes` if only the Hessian section should be refreshed. The GF/PED report includes two diagnostics for this workflow:

- a geometry check reporting raw and optimally aligned RMS/max differences between the frozen `#GIC` reference geometry and the Hessian geometry;
- a frequency check reporting mode-by-mode, RMS and maximum differences between MATRIX GF frequencies and the harmonic frequencies stored by the source adapter.

If `-symmetry-blocks` is requested, GF/PED first checks whether the internal F and G matrices are block diagonal in the stored irrep labels. When off-block couplings are larger than numerical roundoff, GF/PED stops with an error: symmetry blocks from correctly symmetrized GICs must not couple different irreps. Such a failure is therefore a NEO/projector regression to fix upstream, not a GF option to bypass. PED columns are normalized to 100% for every solved mode.

5 Pulay-Style Scaling

GF/PED accepts optional diagonal scaling factors in the GIC basis. Scaling is applied after transformation of the Cartesian Hessian to internal coordinates and before the Wilson GF eigenproblem. A scaling file is line-oriented text or CSV:

```
# selector factor
default 1.000
GIC003 0.980
A1Str0001,0.995
5=1.020
class CH_stretches 0.970 R(1,6)|R(2,7)|R(3,8)
```

Selectors may be:

- `default`, `all` or `*`;
- a one-based index such as `5`;
- a `GICnnn` label;
- an exact GIC name;
- an exact label;
- an unambiguous substring of a name or label.

Inline records are supplied with repeated `-scale` options:

```
python -m matrix gf \
  --xyzin working/molecule.xyzin \
  --fchk working/gauin.fchk \
  --scale-file working/pulay_scale.txt \
  --scale "GIC003=0.980" \
  --scale "default=1.000" \
  --scale-class "CH_stretches:0.970:R(1,6)|R(2,7)|R(3,8)"
```

Class records reuse the semiexperimental-fit idea of named parameter classes, but apply it to diagonal Pulay factors in the frozen GIC basis. A class has a name, one factor and one or more case-insensitive label/name patterns separated by `|`. Patterns in a class may deliberately match several GICs. MATRIX rejects the class if it matches no GIC or if the matched GICs mix coordinate families such as stretches, bends, torsions, out-of-plane coordinates, ring puckering coordinates or special fragment/center coordinates. This prevents a chemically motivated scale factor from silently spanning unrelated internal coordinate types. Scaling factors must be non-negative. Ambiguous or unmatched ordinary selectors remain fatal errors because silent scaling of the wrong coordinate would invalidate the PED interpretation.
Before running GF, the resolved assignments can be inspected without reading an Hessian and without modifying the xyzin file:

```
python -m matrix gf \
  --xyzin working/molecule.xyzin \
  --scale-file working/pulay_scale.txt \
  --scale-class "CH_stretches:0.970:R(1,6)|R(2,7)|R(3,8)" \
  --scale-preview
```

The preview lists every scaling rule, its matched GICs, the detected coordinate family and the final factor assigned to each GIC after later rules override earlier ones.

6 Non-Bonded Hessian Subtraction and Local Models

Before the Cartesian Hessian is transformed to the GIC basis, GF/PED may subtract model electrostatic and van der Waals contributions:

```
python -m matrix gf \
  --fchk working/gauin.fchk \
  --xyzin working/xyzin \
  --subtract-electrostatic \
  --subtract-uff-vdw \
  --nonbonded-14-scale 0.5 \
  --local
```

The charge vector is read from the canonical `#SYNTHONS` section. If Gaussian CM5 charges were imported during LINK preprocessing, those are used. Otherwise the same synthon charge column is present and comes from the ORACLE electronegativity model. GF/PED therefore never reparses Gaussian text for charges.

The van der Waals correction uses ORACLE topology radii and UFF well-depth coefficients where available. Unsupported element pairs fail explicitly rather than silently using zeros.

Both electrostatic and UFF-vdW corrections use the same topological pair policy: 1-2 and 1-3 pairs are excluded, 1-4 pairs are included with `-nonbonded-14-scale` (default 0.5), and pairs beyond 1-4 are included with full weight. Disconnected inter-fragment pairs are treated as beyond 1-4.

The optional `-local` model is applied after this Cartesian subtraction and after transformation to internal coordinates. It keeps diagonal primitive terms and local bonded couplings while zeroing non-local internal force-constant terms.

7 Output Files

Output	Meaning
`gf_ped_report.txt`	Human-readable report for the quick `gf` workflow.
`gic_gf_ped_report.txt`	Human-readable report for the frozen-GIC production workflow.
`gf_manifest.json`	Run manifest for the quick workflow.
`gic_gf_manifest.json`	Run manifest recording FCHK, schema, optional geometry, scaling input and output files.
`*_frequencies.csv`	Mode index and frequency in cm^{-1}.
`*_frequency_comparison.csv`	Optional GF-vs-source frequency comparison written when the source adapter provides harmonic frequencies.
`*_gic_labels.csv`	GIC index, name, irrep, coordinate family, anharmonic class, zero-order model, cross-coupling policy and coordinate label.
`*_ped.csv`	Potential-energy distribution, rows as GICs and columns as modes.
`*_normal_modes.csv`	Internal-coordinate normal-mode vectors.
`*_force_constants.csv`	Internal-coordinate force-constant matrix after optional scaling.
`*_g_matrix.csv`	Wilson G matrix in the same GIC order.
`*_g_inverse.csv`	Equilibrium reference inverse Wilson G^{-1} matrix in the same non-redundant GIC order; large-amplitude DVR setup uses this table for planning, but production grids must recompute $G(Q)$, $g(Q)$, $\det g(Q)$ and the Podolsky/metric-derivative terms.
`*_large_amplitude_coordinates.csv`	Large-amplitude GIC selection with family labels, anharmonic class, local one-coordinate GF frequency and active/excluded status.

Output	Meaning
`*_large_amplitude_blocks.csv`	Mono- and poly-dimensional direct F/G subspace frequencies, anharmonic class and coupling diagnostics, including the combined $\max(F_{\mathrm{rel}}, G_{\mathrm{rel}})$ criterion and the selected G_{SS}^{-1} block used by multidimensional DVR planning at the reference geometry.
`*_large_amplitude_g_inverse_blocks.csv`	Long-form row/column table of the same G_{SS}^{-1} blocks, convenient for numerical audit and downstream readers; values are equilibrium references.
`*_large_amplitude_mode_ped.csv`	Total large-amplitude PED contribution for each GF mode.
`*_large_amplitude_dvr_plan.csv`	DVR-planning large-amplitude table with status, F/G coupling diagnostic, torsional periodicity, minimum position, $(G^{-1})_{ii}$, and one-term Fourier estimate when a 1D treatment is justified.

The text report begins with the Hessian source, coordinate source, point group, symmetrization flag and GIC count. When available, it then prints geometry and frequency diagnostics before the frequencies, GIC labels and PED matrix. In the frozen-GIC workflow, labels and irreps are those stored or evaluated from the `GICForge` schema.

8 GUI Workflow

The advanced GUI exposes GF/PED through the dedicated `GF / PED` window. The window reads a Cartesian Hessian FCHK/FCH file, an optional frozen GIC definition, an optional B-matrix geometry and optional Pulay scaling records. It displays the text report and can export the same CSV tables written by the CLI.

The dashboard also exposes a `GIC Definition / B Matrix / GF-PED` workflow. The intended sequence is:

1. define or load a frozen GIC schema;
2. select the FCHK Hessian;
3. optionally select a current geometry for B evaluation;
4. optionally supply Pulay scaling;
5. run GF/PED and export report/CSV tables.

9 Validation and Failure Policy

GF/PED must fail rather than continue when the mathematical model is invalid. Fatal conditions include:

- FCHK missing masses, Cartesian coordinates or Cartesian force constants;
- non-symmetric Cartesian Hessian;

- schema atom count different from the Hessian input atom count;
- B matrix dimension inconsistent with the Hessian;
- non-positive-definite Wilson G matrix for the selected coordinates;
- non-negligible F/G couplings between different irreps when `-symmetry-blocks` is requested;
- negative Pulay scaling factor;
- ambiguous or unmatched Pulay selector.

A negative vibrational eigenvalue is not a fatal error by itself. It is reported as a negative frequency because it may represent a transition-state direction or an instability in the supplied Hessian/geometry combination.

10 Relation to GICForge, MORPHEUS and VPT2/VCI

`GICForge` owns coordinate construction, pruning and symmetry labels. GF/PED consumes that frozen coordinate model and performs harmonic vibrational linear algebra. MORPHEUS/SEfit consumes the same coordinate model for semiexperimental structure refinement. VPT2/VCI is a separate workflow based on normal-coordinate cubic and quartic force fields and should not be confused with GIC-based Wilson GF/PED.

This separation is intentional. A coordinate definition can be generated once, used to prepare Gaussian input, reused for semiexperimental refinement, and then reused again for GF/PED analysis of the Cartesian Hessian. When the same schema is used throughout, coordinate labels, symmetry species, B matrices and PED assignments remain auditable across modules.